Aguamarina Joyería - Google Ads API Integration Design

1. Tool Overview & Executive Summary

Aguamarina Marketing Platform is an in-house, private marketing automation software built specifically for Aguamarina Joyería. The tool enables internal staff to streamline the creation, configuration, and monitoring of Google Ads marketing campaigns (YouTube Video Ads, Search Ads, Display Ads, and Performance Max) directly from a centralized local dashboard, synchronizing high-value jewelry promotion assets with the official Google Ads account.

2. Technical Architecture & Security Model

• Environment & Technology Stack: Node.js (v20+), Express REST backend, TypeScript, official google-ads-api v25 SDK.
• Hosting Environment: Secure internal localhost application operated exclusively by Aguamarina marketing staff.
• Authentication Model: Google OAuth 2.0 Web Application flow using offline refresh tokens and developer tokens.
• Google Cloud Project ID: 1000741529450 (aguamarina-automation).
• Manager Account (MCC): 720-475-5391 (JOYERIA AGUAMARINA ADMINISTRADOR).
• Client Account (Operating CID): 164-944-3437 (Joyeria Aguamarina).

3. End-to-End Data Flow

Step 1: Campaign Configuration - Staff selects campaign type (Search / Video Ads), inputs daily budget, target locations (San José, Costa Rica), schedule dates, and final destination URL.

Step 2: Client-side Validation - The dashboard validates character lengths (headlines <= 30 chars, descriptions <= 90 chars, call-to-actions) and parses YouTube video IDs before transmission.

Step 3: Secure API Communication - The server securely initiates an authenticated gRPC session using Developer Token and OAuth refresh token via login_customer_id (720-475-5391).

Step 4: Resource Mutate Execution - Executes atomic mutate operations: (1) CampaignBudget, (2) Campaign, (3) AdGroup, and (4) Responsive Search/Video Ad Assets in the target account (164-944-3437).

Step 5: Status & Audit Logging - Live resource identifiers (Resource Names, Campaign IDs) are returned and displayed in the management interface for performance verification.

4. Data Privacy, Storage & Compliance with Google Ads Policies

• Internal Use Exclusivity: The application is strictly for internal business operations of Aguamarina Joyería. It does not sell ads, provide third-party agency services, or offer SaaS access to external entities.
• Data Retention: No third-party ad data or personal identifiable information (PII) is exported, stored, or scraped outside the official Google Ads platform.
• Rate Limits & Quotas: Campaign mutates are triggered on-demand via human user interaction, strictly adhering to Google Ads API rate limits and Minimum Functionality Requirements (MFR).
• Credential Security: OAuth Client secrets and developer tokens are stored securely in protected environment variables (.env) and never exposed publicly.

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